

MOSCATO: Predicting Multiple Object State Change through Actions

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Abstract

We introduce *MOSCATO*: a new benchmark for predicting the evolving states of multiple objects through long procedural videos with multiple actions. While prior work in object state prediction has typically focused on a single object undergoing one or a few state changes, real-world tasks require tracking many objects whose states evolve over multiple actions. Given the high cost of gathering framewise object-state labels for many videos, we develop a weakly-supervised multiple object state prediction framework, which only uses action labels during training. Specifically, we propose a novel Pseudo-Label Acquisition (PLA) pipeline that integrates large language models, vision-language models, and action segment annotations to generate fine-grained, per-frame object-state pseudo-labels for training a Multiple Object State Prediction (MOSP) network. We further devise a State-Action Interaction (SAI) module that explicitly models the correlations between actions and object states, thereby improving MOSP. To facilitate comprehensive evaluation, we create the *MOSCATO* benchmark by augmenting three egocentric video datasets with framewise object-state annotations. Experiments show that our multi-stage pseudo-labeling approach and SAI module significantly boost performance over zero-shot VLM baselines and naive extensions of existing methods, underscoring the importance of holistic action-state modeling for fine-grained procedural video understanding.¹

1. Introduction

Long-form video understanding, particularly for procedural tasks, has recently achieved significant attention due to its application in assistive technologies and robotics [14, 27, 28, 36, 52, 61]. The majority of existing research has focused on coarse-level video understanding, such as action recognition [18, 23, 39, 63, 67, 75], temporal action segmentation (TAS) [5, 13, 16, 31–33, 51, 53, 58, 68], and action anticipation [19, 20, 38, 50, 71], as well as detecting,

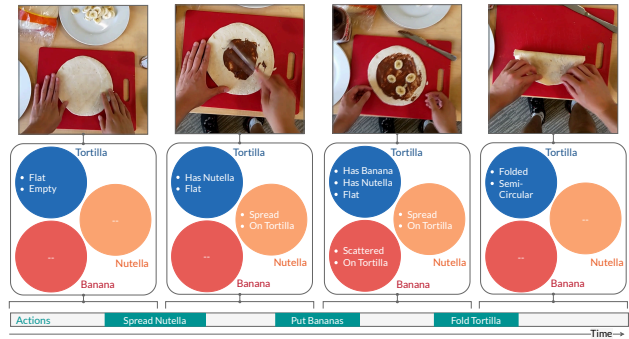


Figure 1. We study the new problem of *multiple object state change prediction* in long videos, in which relevant objects to the actions being performed may undergo several state changes.

segmenting and tracking relevant objects [54, 72, 76]. Yet, fine-grained procedural video understanding has remained a challenging and an under-explored problem [15, 26, 27].

Specifically, in long procedural task videos, the states of multiple objects change as we progress in a task, see Fig. 1. Each action/step often changes the state of several objects (e.g., *spreading Nutella* changes the states of both tortilla and Nutella) and the state of an object often changes over several actions (e.g., the state of tortilla moves from *empty* to *has Nutella* to *has banana* to *folded*). Being able to understand *which objects* undergo state changes as well as *when and how* their states change helps in better and more fine-grained *temporal and spatial* reasoning about videos. It allows better understanding of the actions, progress and error (e.g., if the tortilla never reaches the state *has banana* before *folded*, there is a missing *put bananas* step) and provides explainability about these predictions. In particular, it enables task assistants to obtain spatial and temporal awareness of step-relevant parts of the scene and video for knowing when and how to intervene.

SOTA Limitations. To the best of our knowledge, the problem of predicting (multiple) state changes of multiple objects in a video has not been studied and there is no dataset for evaluation of this task. Prior works in object state prediction have mainly focused on single-object scenarios, such as detecting and localizing a single state change for

¹Code and data is publicly available at https://github.com/eelhami/iccv25_moscato.

only one object in a video [4, 55, 56]. While more recent methods have considered multiple state changes [59] or one state change with an open-world object vocabulary [66], they still handle a *single* object per video. However, real-world procedures typically involve *multiple* objects that undergo state changes when performing *multiple* actions in a video (see Fig. 1). Yet, it is very costly and challenging to densely annotate multiple possible states of many objects across many videos (currently there is no annotated dataset).

On the other hand, there is a tight relationship between actions and states: knowing the action provides information about the state changes, and knowing state changes can indicate the action. This suggests that for effective video understanding, we should learn to predict both actions and state changes while capturing and leveraging their interactions. In fact, while some studies have leveraged object cues to improve action recognition, action anticipation or temporal action segmentation [69, 70, 74], the opposite direction (i.e., exploiting temporal action segmentation to improve object state prediction), remains an open problem.

Paper Contributions. We study the new problem of *multiple object state prediction* (MOSP) in long procedural videos and develop an efficient framework to address it without using annotated state changes in videos. As shown in Fig. 1, our goal is to detect and track the state of *every* task-relevant object (undergoing state changes) in a video. Motivated by task assistant and robotic applications, we focus on egocentric videos [12, 22], where the first-person view naturally follows the user’s interactions with objects and involves dynamic and frequent object state changes.

We address several major challenges. First, to learn without using annotated object states in videos, we develop a *Pseudo-Label Acquisition* (PLA) pipeline that leverages action labels, large language models (LLMs) and vision-language models (VLMs) to automatically generate object-state pseudo-labels for training videos. While PLA provides state descriptions of relevant objects before and after each action, we do not have their temporal localizations to train a MOSP model. Therefore, we propose a point of change (POC) estimation by leveraging VLMs that enables localizing the before-action and after-action states of objects. We then study a joint multiple object state prediction and temporal action segmentation (TAS) architecture in which the two interact, and particularly, TAS informs and improves MOSP. To do so, we propose a *State–Action Interaction* (SAI) module that takes advantage of the intrinsic correlation between action labels and object state labels.

To validate our method, given the lack of a dataset with annotated state changes for multiple objects, we create the **MOSCATO** (*Multiple Object State ChAnge ThrougH actiOns*) benchmark (see Table 1), where we extend three existing egocentric TAS datasets with extensive per-frame object state annotations.

2. Related Works

Object State Prediction. Early research studied object states primarily in images, focusing on transformations and compositional relations [24]. Subsequent works extended to videos, learning to discover and track object states [2, 4, 17, 44, 48, 62]. Other approaches investigated compositional generalization in zero-shot settings, but mainly in images or with a limited scope of attributes [29, 34, 37, 40, 41, 45, 46, 49]. More recent video-based methods began classifying or localizing object state changes alongside actions [35, 55–57], often focusing on a single object undergoing one or a few state transitions [59, 66]. In contrast, we propose a more fine-grained multiple object state prediction task, tracking *all* relevant objects and their evolving states across the entire sequence of actions.

Datasets on Object States. Several prior datasets include some form of object state annotations, but they often have significant limitations. Image-based datasets [21, 49] cover only a limited number of object or state categories. Video datasets focus on a single object undergoing one or a few transitions [55, 56, 66], or place constraints on how often the state can change within a single video [59]. Meanwhile, [35] and other recent efforts [43, 57] target specialized tasks (e.g., state-focused captioning or anticipation). Consequently, none of these datasets are well suited for modeling *multi-object* state changes throughout longer procedural videos. Our work addresses this gap by enriching existing egocentric video datasets with fine-grained state annotations for *all* relevant objects and creating a benchmark to evaluate multiple object state prediction.

LLM/VLM Knowledge Utilization. Large Language Models and Vision Language Models have demonstrated robust reasoning and comprehension skills across tasks such as recognition, localization, object detection, and visual question answering [3, 9, 10, 60, 64, 72, 73]. Despite their strong performance, they can struggle with certain nuances such as intricate object relations or subtle attribute changes [7, 42]. In our framework, we harness the general world knowledge of these models for pseudo-label generation and correlation discovery, then complement them with post-processing and refinement steps to address their failure modes and produce accurate multi-object state annotations.

3. Multiple Object State Prediction (MOSP)

Problem Statement. Assume we have a sequence of T video frames $\mathcal{V} = (v^{(1)}, v^{(2)}, \dots, v^{(T)})$, where $v^{(t)} \in \mathbb{R}^{H \times W \times 3}$ denotes the RGB frame at time t with the size $H \times W$. The goal of MOSP is to predict the sequence of object states $\mathcal{Y} = (\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(T)})$, where $\mathbf{y}^{(t)} \in \{0, 1\}^{O \times S}$ is the predicted object-state label matrix with $y_{os}^{(t)}$ denoting whether object o has state s at frame t . Here, O is the total number of objects in the dataset, and S is the

Datasets		Egocentric	# Obj	# St	# Vid	# StC/Obj	# Obj/Vid	Avg. Video Duration(s)	GT/Pseudo
ChangeIt (Training) [55]		×	42	27	34428	1	1	276.0	Pseudo
ChangeIt (Evaluation) [55]		×	42	27	667	1	1	276.0	GT
HowToChange (Training) [66]		×	122	20	36075	1	1	41.51	Pseudo
HowToChange (Evaluation) [66]		×	134	20	5424	1	1	41.51	GT
MOST (Training)[59]		×	6	60	10749	> 1	1	336.9	Pseudo
MOST (Evaluation) [59]		×	6	60	61	> 1	1	156.9	GT
MOSCATO (Training)	CMU	✓	39	77	129	> 1	> 1	353.3	Pseudo
	EGTEA	✓	74	79	61	> 1	> 1	1083.5	Pseudo
	EgoPER	✓	77	210	163	> 1	> 1	150.3	Pseudo
MOSCATO (Evaluation)	CMU	✓	39	77	36	> 1	> 1	358.3	GT
	EGTEA	✓	74	79	25	> 1	> 1	1781.5	GT
	EgoPER	✓	77	210	50	> 1	> 1	148.1	GT

Table 1. Comparison of MOSCATO to previous object state change video datasets. # Obj and # St denote the number of objects and states, respectively, in the object vocabulary. # Vid represents the number of videos. # OSC/Vid is the average number of object state changes per video. # StC/Obj is the number of state changes for an object in a video. # Obj/Vid denotes the number of objects that undergo state change in a video. GT/Pseudo determines if that subset is manually (GT) or automatically (Pseudo) labeled.

total number of possible states for these objects. Therefore, we study a closed-vocabulary setting in the paper. Given the high cost of gathering ground-truth object-state labels for videos, we assume we do not have ground-truth object-state labels for training videos. Instead for every training video V , we have its action labels, $(z^{(1)}, z^{(2)}, \dots, z^{(T)})$, where $z^{(t)} \in \{1, \dots, C\}$ denotes the action class of frame t and C is number of action classes.

Overview. Our setting can be considered as a *weakly-supervised* multiple object state prediction problem, since we do not have object-state labels during training and instead use action labels. To tackle the problem, we develop a framework that consists of several components, as illustrated in Fig. 2. First, we use a MOSP network for multiple object state prediction. Given the lack of ground-truth object-state labels, we design a Pseudo-Label Acquisition (PLA) module, which leverages LLMs to obtain object and state vocabulary and VLMs to assign the states to specific frames using a point of change estimation method, which we propose. Therefore, using the output of PLA, we can supervise training of the MOSP network. To leverage the correlations between actions and states, we use conventional TAS models to predict action labels and design a novel State-Action Interaction (SAI) module to leverage these correlations for improving object state predictions.

3.1. Pseudo-Label Acquisition (PLA)

TAS datasets [6, 11, 25, 27] generally provide action segment annotations (these can also be derived from narrations without access to explicit annotations). We leverage these to build framewise object-state pseudo labels for training. We propose a multi-stage pipeline (Fig. 3) that utilizes LLMs to progressively generate and refine the object state labels. We further use vision-based reasoning to localize segments associated with these states.

Stage 1: Object State Description via LLM. Given the ac-

tion label of each segment, we prompt an LLM, here GPT-4o [3], to 1) identify the objects likely involved in the action and 2) describe how the appearance or state of each object changes after the completion of the action. Since the order of actions of a task can vary across videos (e.g., pouring water before adding a teabag or vice versa), the sequence of states an object undergoes may differ across videos of the same task. Thus, we additionally feed the previous actions and object states to the LLM to provide a history context.

Stage 2: State Description Summarization. The previous stage produces free-form sentences describing the states of objects. To summarize these descriptions, we prompt the LLM to extract adjectives or short phrases that capture the essential states. We include short phrases because some states, such as “in a mug” and “vegetable covered”, cannot be described using only adjectives. Meanwhile, we already obtained a list of objects from stage one. Because the same object or state can appear under various names (e.g., “mug” and “cup” or “chopped” and “cut”), we then apply k -means clustering on the text embeddings² for both objects and states to merge any redundant or synonymous labels into a vocabulary of distinct object and state categories. We use the Elbow method and Sum of Squared Errors to guide the choice of k .

Stage 3: Object State Refinement. Even with historical states provided, the LLM may occasionally omit valid states or fail to preserve ongoing attributes. For instance, if vegetables are chopped and then put on a bun, the LLM might only mention “on buns” for the vegetables and miss the “chopped” state. Given that we have collected a list of all possible states for each object from the LLM’s responses over all videos, we refine the states by asking the LLM to check if any other state applies to the object. Next, we ask the LLM to improve the tracking of states by asking if any of the object states before the action would still be applica-

²We use the `text-embedding-3-large` model in GPT API.

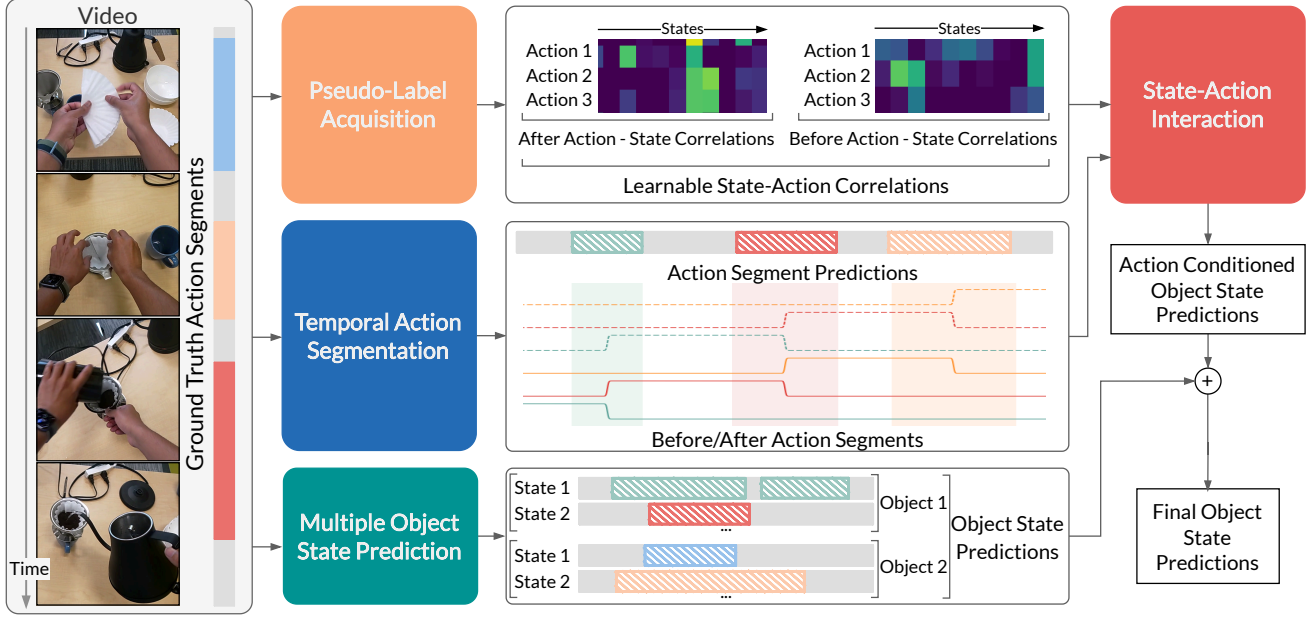


Figure 2. Overview of our proposed framework for the task of multiple object state prediction.

ble after the action.

Stage 4: Point-of-Change (POC) Estimation. So far, we obtained state descriptions for objects in each action segment, however, without temporal localization, which is necessary to train a MOSP model. Thus, at this stage, we assign frame-wise object state labels to the training videos. To do so, we localize the *Point-of-Change* (POC), hence, divide each action segment into segments for *before-action* state and *after-action* state. In each action segment, POC is where the state transition most likely occurs. In the paper, we assume that *all* involved objects in an action switch states at the same moment (an approximation that simplifies the process). Our experiments also show that it yields better results than attempting to track separate POCs per object.

Let the set of involved objects in an action be $O = \{o_1, o_2, \dots, o_K\}$. Each object o_i has a *before-action* state $S_{o_i}^B$ (the state of objects from the previous POC to the current one) and an *after-action* state $S_{o_i}^A$. We remove the states common to both $S_{o_i}^B$ and $S_{o_i}^A$ since they do not help distinguish the *before* vs. *after* transition,

$$\tilde{S}_{o_i}^B = S_{o_i}^B \setminus S_{o_i}^A, \quad \tilde{S}_{o_i}^A = S_{o_i}^A \setminus S_{o_i}^B. \quad (1)$$

Using a pre-trained CLIP [47] model, we compute the cosine similarity between the visual embeddings of video frames and the text embeddings of object states. For object o_i , we then define the average similarity of frame t to *before-action* (B) and *after-action* (A) states as

$$\omega_{o_i}^x(t) = \frac{1}{|\tilde{S}_{o_i}^x|} \sum_{s \in \tilde{S}_{o_i}^x} \text{Sim}(E^{img}(v^{(t)}), E^{text}(s)), \quad (2)$$

where $x \in \{B, A\}$, $v^{(t)}$ is the RGB image at frame t , E^{img} and E^{text} are the image and text encoders of CLIP. We then

compute the POC score $\gamma(t)$ using all K objects involved in the action as,

$$\gamma(t) = \frac{1}{K} \sum_{i=1}^K \frac{\omega_{o_i}^A(t)}{\omega_{o_i}^B(t)}, \quad (3)$$

which measures how much closer frame t is to the *after-action* states than to the *before-action* states for *all* objects in the action. For each action segment, we take the frame where this ratio exhibits the sharpest increase as the estimated POC, and assign *before-action* and *after-action* labels to frames accordingly.

Stage 5: Object Visibility Detection. Finally, we employ a pre-trained object detection model (GLIP [72]), prompting it with the object vocabulary for each task. This allows us to determine in which frames each object actually appears. We then assign final pseudo-labels only to frames where the object is visibly present.

All complete prompts are included in the supplementary material. Through PLA, we obtain per-frame pseudo object-state labels for the training videos, which we use to train a MOSP model, which we discuss next.

3.2. Multiple Object State Prediction (MOSP) Model

Using frame-wise object-state pseudo-labels, obtained from PLA, we train a MOSP model (Fig. 2). Following previous works [59], we adopt a TAS architecture (MSTCN++ [30]) as our MOSP model to capture long-range temporal dependencies. Unlike state prediction methods for a single object [55, 56, 66], our multi-object scenario requires classifying a large number of possible object and state categories for each frame in the video. Modeling the temporal evo-

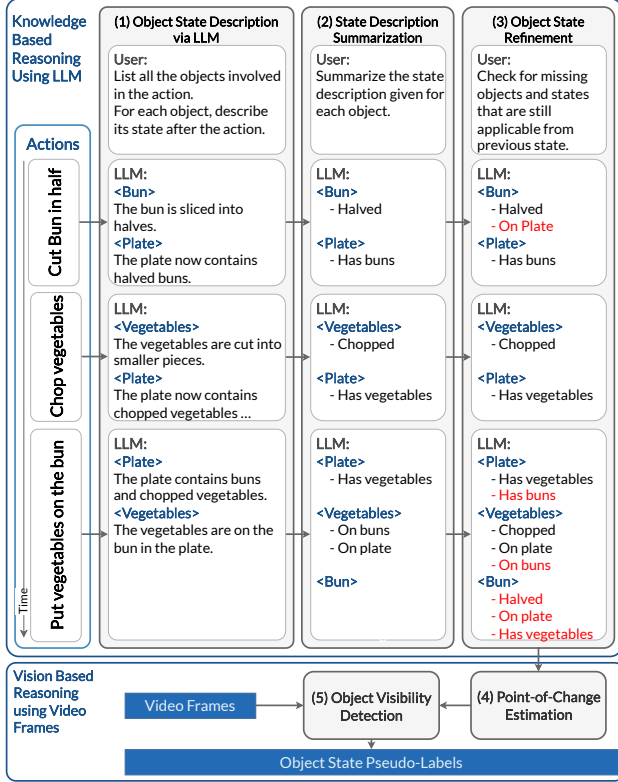


Figure 3. Overview of our proposed object-state pseudo-labeling process: We first use LLM-based reasoning with ground-truth action annotations to extract object vocabulary and object states. Then, we use vision-based reasoning on frames, aggregating the information into final object-state pseudo-labels. In the *Object State Refinement* stage, red highlights object states that the LLM may miss in the *State Description Summarization* stage.

lution of these object states can be viewed as a multi-label classification problem, where each video frame may contain multiple objects, each in potentially different states. To handle the class imbalance in these multi-label outputs, we use a weighted binary cross-entropy loss,

$$\mathcal{L}_{\text{MOSP}} = \sum_{t,o,s} \left[w_p y_{os}^{(t)} \log p_{os}^{(t)} + (1 - y_{os}^{(t)}) \log(1 - p_{os}^{(t)}) \right], \quad (4)$$

where $p_{os}^{(t)}$ and $y_{os}^{(t)}$ are the prediction and pseudo-label for object o and state s at frame t , and w_p is a weight factor reflecting class imbalance.

3.3. State-Action Interaction (SAI) Model

Actions and object states are closely intertwined: i) only specific actions can cause certain state changes (e.g., “hold cup” cannot change the state of a cup from *empty* to *full*); ii) actions affect only specific objects (e.g., “pour water into mug” does not change the state of the cutting board); iii) some actions require objects to be in a specific state (e.g., a hot dog must be *cut* before being spread on pizza).

To leverage dependencies between actions and object states, we use a TAS model to predict framewise action

labels and propose a *State-Action Interaction (SAI)* module to refine the outputs of the MOSP branch using action predictions from the TAS branch. Following previous works [16, 30, 32], we use a cross-entropy loss and a smoothing loss to train the TAS model using the groundtruth action labels, i.e., $\mathcal{L}_{\text{TAS}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{smooth}}$. Next, we discuss the details of using action-state correlations in SAI for improving the MOSP.

SAI Formulation. Our goal is to leverage action predictions from TAS to obtain object-state predictions, which we then use as *complementary pseudo-labels for training* the MOSP and as *complementary object-state predictions during inference*.

Notice that a video can be segmented based on actions or object states. Let $\mathcal{A} = (a_1, a_2, \dots, a_N)$ denote a sequence of N action segments in a video, where the temporal boundary of the action segment a_i is denoted by $[t_i^{\text{start}}, t_i^{\text{end}}]$. Using $\mathcal{A}$, we can build a sequence of *object state segments* as $\mathcal{U} = (u_0, u_1, \dots, u_N)$. Here, each u_i spans from the midpoint of action segment a_i to the midpoint of action segment a_{i+1} , which we denote by the interval $[m_i, m_{i+1}]$. We assume that object states remain constant in each u_i .

Our goal is to obtain $P(u_i)$, the likelihood of each state segment u_i using the TAS output, in order to further supervise the MOSP and improve its predictions (see below). Therefore, we consider u_i ’s adjacent actions a_i and a_{i+1} . We use $P_A(u_i|a_i)$ and $P_B(u_i|a_{i+1})$ to denote the probabilities of u_i being immediately *after* a_i and *before* a_{i+1} , respectively. Meanwhile, we compute $P_A(a_i)$ and $P_B(a_{i+1})$ using TAS predictions, capturing the confidence of predictions for a_i and a_{i+1} . We then approximate

$$P(u_i) \approx f\left(P_A(u_i|a_i)P_A(a_i), P_B(u_i|a_{i+1})P_B(a_{i+1})\right), \quad (5)$$

where the learnable function $f(\cdot)$ is realized by a 1D convolution and fuses these conditional probabilities into an estimate of u_i . We use Algorithm 1 to precompute empirical values for $P_A(u|a)$ and $P_B(u|a)$ using training videos.

Once $P(u_i)$ is determined, each frame $t \in [m_i, m_{i+1}]$ within u_i adopts this probability for its corresponding object-state pairs. Let (o, s) be one such object-state combination; we define

$$q_{os}^{(t)} = P(u_i), \quad \forall t \in [m_i, m_{i+1}]. \quad (6)$$

We then form the loss $\mathcal{L}_{\text{SAI}}$ using the weighted binary cross entropy similar to Eq. (4) applied to $q_{os}^{(t)}$.

3.4. Training and Inference

During training, we learn the MOSP and TAS parameters as well as SAI (f in Eq. (5)) by using the training loss

$$\mathcal{L} = \mathcal{L}_{\text{MOSP}} + \mathcal{L}_{\text{SAI}} + \mathcal{L}_{\text{TAS}}. \quad (7)$$

Algorithm 1 Obtaining State-Action Correlations from Pseudo-Labels. The pseudocode is shown for a single video. In practice, the same process is repeated for all videos in the training set.

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1: Input: object-state pseudo labels  $\mathcal{Y}$ ; action sequence  $\mathcal{A}$  with
   associated start, end, and middle times  $\{t_i^{\text{start}}, t_i^{\text{end}}, m_i\}_{i=1}^N$ .
2: Initialize counts  $n_B(o, s, a) \leftarrow 0, n_A(o, s, a) \leftarrow 0$ 
3: for action  $a_i$  in  $\mathcal{A}$  do
4:   for object  $o$  in  $\{1, 2, \dots, O\}$  do
5:     for state  $s$  in  $\{1, 2, \dots, S\}$  do
6:        $n_B(o, s, a_i) + = \sum_{t \in [t_i^{\text{start}}, m_i]} y_{os}^{(t)}$ 
7:        $n_A(o, s, a_i) + = \sum_{t \in [m_i, t_i^{\text{end}}]} y_{os}^{(t)}$ 
8:     end for
9:   end for
10: end for
11:  $P_A(u|a) = \frac{n_A(o, s, a)}{\sum_{o, s} n_A(o, s, a)}, \quad \forall o, s, a \text{ and } u = (o, s)$ 
12:  $P_B(u|a) = \frac{n_B(o, s, a)}{\sum_{o, s} n_B(o, s, a)}, \quad \forall o, s, a \text{ and } u = (o, s)$ 
13: Output:  $P_B(u|a)$  and  $P_A(u|a)$ .

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For inference, we obtain $p_{os}^{(t)}$ from the MOSP branch and $q_{os}^{(t)}$ from the SAI branch. The final MOSP prediction for object states is a weighted sum of the raw MOSP output and the refined SAI output,

$$p_{os}^{(t), \text{Final}} = (1 - \beta) p_{os}^{(t)} + \beta q_{os}^{(t)}, \quad (8)$$

where the hyperparameter β controls the influence of SAI and $p_{os}^{(t), \text{Final}}$ is the final object state probabilities predicted by our method. Ablation study on the hyperparameter β is included in the supplementary materials. Empirically, modeling action–state relationships yields significant gains in MOSP performance.

4. MOSCATO Dataset

We introduce **MOSCATO** (Multiple Object State ChAnge Through actiOns): a new dataset that captures *multiple* object state changes throughout a sequence of actions. As illustrated in Fig. 1, MOSCATO goes beyond merely tracking one primary object of interest during actions; it also documents other objects that may experience significant changes. We argue that the states of these additional objects are often just as important as those of the primary object, providing a richer context.

Data Collection. We build MOSCATO by leveraging three egocentric procedural task datasets: CMU [6], EGTEA [6], and EgoPER [27], each featuring various cooking tasks and object state changes. On these datasets, the videos differ in how tasks are ordered, forcing models to learn from the history of actions and object states rather than relying on a fixed sequence of actions or object states. Such variability ensures a more robust evaluation of the MOSP models. CMU, EGTEA, and EgoPER contain 165, 86, and 386 videos, respectively. We select 36 (CMU), 25 (EGTEA),

and 50 (EgoPER³) videos as test sets, and manually annotate them with ground-truth object state labels. The remaining videos form the training sets, for which we generate pseudo-labels using our approach in Sec. 3.1.

Data Annotation. For each dataset, we first collect a vocabulary of objects and a list of corresponding states by our proposed PLA pipeline (see Sec. 3.1). For the test splits, we collect manual annotations using LabelBox [1]. Each annotator first goes through the object list and annotates the frames in which each object is visible. Annotators then select the appropriate state for each visible object from a pre-defined list of possible states for each object.

Dataset Statistics. Table 1 compares MOSCATO to prior datasets. While ChangeIt [55] and HowToChange [66] focus on a single state change per video, Ego4D-OSCA [35] or MOST [59] track multiple state changes, but only for a single object. MOSCATO captures state changes across *all* within an entire video offering a broader and more complete perspective. While ChangeIt [55] and HowToChange [66] contain more manually annotated videos, the annotations are high-level and are categorized into three categories: “initial state”, “action”, and “final state”. Similarly, although Ego4D-OSCA [35] provides a large number of annotated videos, the taxonomy of state changes consists of only a few general categories (e.g., “Remove,” “Activate,” “Deform”). In contrast, MOSCATO offers a richer, more comprehensive view of how multiple objects undergo state changes over the course of complex, real-world tasks.

5. Experiments

5.1. Experimental Settings

Evaluation. Following MOST [59], we measure frame-level performance using *F1-max* and *mean Average Precision (mAP)*. *F1-max* is defined as the maximum F1 score across different thresholds (over [0.1, 0.2, ..., 0.9]), making it comparable to uncalibrated zero-shot models. Unlike MOST, which applies separate thresholds for each state category, we adopt a single threshold for all object states due to the large number of object–state pairs in our benchmark. We further report three segment-level metrics, *F1@0.1*, *F1@0.25*, and *F1@0.50*, mirroring standard temporal action segmentation benchmarks. We choose these metrics to evaluate the temporal precision predictions of our model. For all F1 scores, we compute true positives, false positives, and false negatives across the entire test set before calculating the final F1. For EgoPER, we train and evaluate the models of each task separately and report the average performance. For CMU and EGTEA, we train one model per dataset. We do not use existing object state change datasets (e.g., ChangeIt [55], HowToChange [66], MOST [59]) because they lack multi-object annotations.

³For EgoPER, we only use the 213 normal videos, which do not have mistakes, to form the training and test sets.

		MOSP					TAS				
		Frame-wise		Segment-wise							
		F1-Max	mAP	F1@0.1	F1@0.25	F1@0.5	Edit	Acc	F1@0.1	F1@0.25	F1@0.5
EgoPER	CLIP [47]	27.11	0.18	0.98	0.68	0.28	4.92	16.49	3.29	1.40	0.72
	InternVideo2 [65]	27.05	0.20	8.77	6.78	3.76	15.57	11.84	8.05	6.04	2.40
	Pseudo-Labels	38.39	-	39.77	35.55	28.73	-	-	-	-	-
	Ours (w/o SAI)	<u>49.99</u>	35.58	<u>54.95</u>	<u>46.31</u>	<u>41.07</u>	<u>90.78</u>	81.82	92.43	90.54	82.58
	Ours (w/ SAI)	52.83	<u>35.06</u>	58.45	48.84	43.42	91.24	<u>81.48</u>	<u>91.95</u>	<u>89.30</u>	<u>81.66</u>
CMU	CLIP [47]	12.15	0.05	0.68	0.31	0.11	3.03	17.4	1.47	1.04	0.43
	InternVideo2 [65]	9.36	0.07	3.19	2.39	1.19	13.88	34.84	10.12	6.4	2.69
	Pseudo-Labels	17.96	-	24.65	21.91	16.43	-	-	-	-	-
	Ours (w/o SAI)	<u>30.55</u>	<u>11.23</u>	<u>43.67</u>	38.16	<u>26.75</u>	<u>47.77</u>	83.85	<u>49.99</u>	<u>47.47</u>	<u>38.23</u>
	Ours (w/ SAI)	35.61	11.50	43.90	<u>37.60</u>	29.99	49.75	<u>83.69</u>	51.46	49.12	39.76

Table 2. Multiple Object State Prediction and Temporal Action Segmentation results on MOSCATO-EgoPER and MOSCATO-CMU. SAI stands for State-Action Interaction module. We highlight the **best** and second best results.

Setup		State Prediction			
Refinement	F1-Max	F1@0.1	F1@0.25	F1@0.5	
$\times$	7.49	13.04	4.34	2.17	
$\checkmark$	17.96	24.65	21.91	16.43	
CLIP	12.15	0.68	0.31	0.11	

Table 3. Ablation studies on pseudo-labeling stages for CMU.

Baselines. We evaluate two categories of models: *zero-shot* and *trained*. For the *zero-shot* setting, we use an image-based VLM, CLIP (ViT-L/14) [47], and a video-based VLM, InternVideo2 [65]. Specifically, we convert each object-state pair into a text prompt of the form “*a/an {adjective} {object} in the image*” or “*a/an {object} {state phrase} in the image*”, then compute cosine similarity between the frame/video embeddings and text embeddings. If the similarity exceeds a threshold, we predict that object-state combination for the frame. After experimenting with multiple prompt templates, we report results from the best-performing variant. Additionally, we include a *Pseudo-Labels* method, which applies the PLA pipeline to the test set to generate pseudo object state labels. We note that this *Pseudo-Labels* method uses the ground-truth action labels of the testing videos while the others do not. For the *trained* setting, we use MSTCN++ [30] as backbone for MOSP and TAS. Ablation studies performed on the architecture of the backbone is included in the supplementary materials. We compare the performance of using only MOSP and MOSP+SAI to evaluate the effectiveness of our proposed SAI module.

Implementation Details. We extract 2048-dimensional features from an I3D model [8] pre-trained on Kinetics [8]. For MSTCN++, we follow the default configurations provided in the original code, but replace the softmax layer with a sigmoid function to accommodate multi-label classification. The network is composed of four stages, each containing ten dilated convolution layers. For training, we use the Adam optimizer with a learning rate of 1×10^{-5} , and set the class weight w_p to 5.0 to mitigate class imbalance.

5.2. Experiment Results

Main Results. Table 2 shows our evaluations on two subsets of the MOSCATO benchmark (EgoPER and CMU). For CLIP and InternVideo2, we filter out object-state pairs that never appear in the dataset to avoid impossible predictions. However, both methods still perform poorly on MOSP, highlighting the difficulty of recognizing object states without dedicated pseudo-labeling and training. In contrast, our automatically generated pseudo-labels substantially outperform these VLM-based baselines, underscoring the effectiveness of our PLA pipeline (which includes multi-stage refinement, point-of-change estimation, and object visibility detection). Building on these pseudo-labels, our final approach (w/ SAI) achieves consistent gains on EgoPER and CMU, improving F1-max by 14.44% on EgoPER and 17.65% on CMU. It also significantly improves the segment-wise metrics (*e.g.*, F1@0.5) by 14.69% and 13.56%, respectively. Our method without SAI reveals that SAI contributes significantly to both frame-level and segment-level performance, affirming that explicit action-state modeling is critical for MSOP. In particular, SAI increases F1-max by 2.84% on EgoPER and 5.06% on CMU. We also see notable improvements in segment-wise metrics, where we achieve 2.35% and 3.24% boosts in F1@0.5 on EgoPER and CMU respectively.

Although TAS is not our primary focus, from Table 2 we see that TAS metrics remain competitive or even improve slightly under our unified framework. In particular, while on EgoPER, Edit and F1@0.25 for TAS decrease by less than 0.5% when using SAI, on CMU, they increase by at least 1.6% when using SAI.

Table 4 shows our evaluations on the EGTEA subset of the MOSCATO benchmark. EGTEA is particularly challenging due to its diverse tasks, actions, and environments. This dataset differs notably from EgoPER and CMU: it contains longer videos (30 minutes in EGTEA vs. 10 minutes CMU) and includes many background actions such

	Frame-wise		Segment-wise		
	F1-Max	mAP	F1@0.1	F1@0.25	F1@0.5
CLIP [47]	5.97	0.01	0.09	0.03	0.00
InternVideo2 [65]	5.66	0.01	0.83	0.66	0.29
Pseudo-Labels	11.38	-	24.24	15.15	9.09
Ours (w/o SAI)	12.85	2.57	1.24	0.0	0.0
Ours (w/ SAI)	8.04	2.78	10.52	6.89	6.45

Table 4. Multiple Object State Prediction and Temporal Action Segmentation results on MOSCATO-EGTEA for the **MOSP** task. Please refer to supplementary materials for TAS results.

as putting away dishes in a make a sandwich task. These factors make both TAS and object-state prediction more challenging and cause all methods to struggle. CLIP and InternVideo2 yield low scores, suggesting that merely relying on VLMs is insufficient. By contrast, our Pseudo-Labels perform significantly better, especially on segment-wise metrics. Interestingly, Ours (w/o SAI) achieves the highest frame-wise F1-Max but underperforms segment-wise, whereas Ours (w/ SAI) improves mAP and outperforms in segment-wise F1, underscoring the value of explicit action-state modeling. Notably, the Pseudo-Labels still lead in segment-wise metrics, likely because they have access to ground-truth action labels, whereas our methods use predicted ones.

Qualitative Results. Fig. 4 compares our proposed method to the InternVideo2 baseline and manual annotations for a video from the test set. Due to space limitations, we have included only object state predictions from one object. For almost all objects depicted, InternVideo2 predictions do not capture any changes in the objects’ states and predict many false positive predictions both frame-wise and segment-wise. This indicates the lack of ability to reason temporally over video. Also, video features from InternVideo2 are incapable of distinguishing objects in various states. However, our method is sensitive to the states of the objects. We observe that as actions unfold, our model can keep track of the states of objects.

In Fig. 5, we show the *State-Action Correlations* ($P_B(u|a)$ and $P_A(u|a)$) calculated in our SAI branch. We observe these correlations are consistent with our expectations of object-state and action interactions. For example, we see the correlation between action “Put banana slices on tortilla” and “Tortilla/Topped with bananas” increases in After-Action State-Action Correlations.

Ablation Studies on PLA. Table 3 examines the contribution of components in our PLA pipeline. Specifically, we omit the Stage 3 (*object state refinement*) to measure its individual impact. We observe that adding the refinement stage improves segment-wise F1@0.5 by 14.26% and F1-Max by 10.47%, confirming that refinement substantially improves the quality of pseudo labels, highlighting the importance of refinement to correct omissions. Additionally, in previous work [66], CLIP was used for assigning pseudo-labels to frames. We also compare with CLIP as a pseudo-

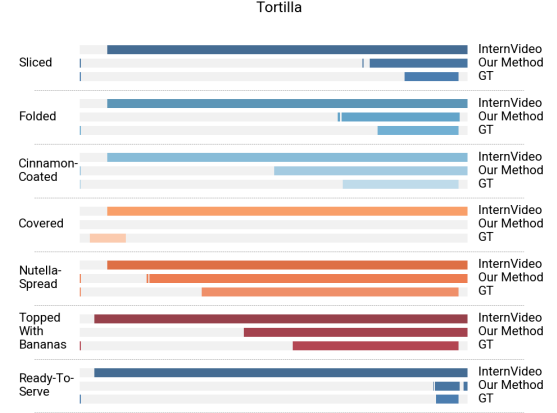


Figure 4. Qualitative comparison of InternVideo, MOSP, and ground-truth annotations for a video from the test set of EgoPER for the object “Tortilla” from the task “Making Quesadilla”. MOSP segments better align with ground truth.

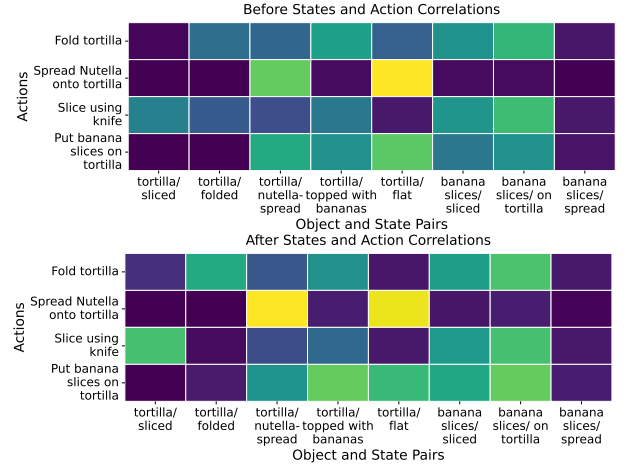


Figure 5. An example for *State-Action Correlations* computed in Alg. 1. The figure illustrates the changes in correlations as an object transition from *before-action* to *after-action*. For clarity in presentation, we only show a few samples of the object-state pairs.

labeling approach. We observe that while CLIP might be effective for short video segments and one object, it cannot effectively track multiple object states.

6. Conclusions

We investigated multiple object state prediction in a weakly-supervised setting. Unlike prior works that focus on a single object, our approach captures multi-object state transitions. We developed a framework that trains a state prediction model using object-state pseudo-labels obtained by a novel approach that leverages LLMs and VLMs. Our framework also leverages correlations between actions and states using a novel state-action interaction module for improved learning and inference. We introduced the new evaluation benchmark of MOSCATO and, by experiments, show that our proposed framework improves over SOTA baselines.

Acknowledgement

This work was funded, in part, by ARPA-H (1AY2AX000062), DARPA PTG (HR00112220001), NSF (IIS-2115110), ONR (N000142512287) and ARO (W911NF2110276). The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the US Government.

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